Ilya Volkov

Backend / DevOps Engineer · Python, Kubernetes

Belgrade, Serbia · remote or hybrid in Belgrade

iillyyaa159@gmail.com Telegram @devtemp github.com/iillyyaa1997 cv.unicorns.place

SUMMARY

Backend / DevOps Engineer with 7 years of experience delivering Python and .NET services, from development through production deployment on Kubernetes, monitoring and ongoing maintenance. Developed features for an online proctoring platform's Django backend, built new APIs in .NET, managed product releases and packaged the product into an on-prem installer, also contributing to the installer itself. Built AI tools on Claude Code that the team adopted. Use Go and Swift in personal projects, choosing the language to suit the task

EXPERIENCE

Constructor Tech (formerly Examus, Alemira) — Backend / DevOps Engineer `Nov 2020 – Oct 2026`

Online proctoring platform · team of 10–20

After Alemira acquired Examus, I moved to Alemira, which was later renamed Constructor Tech

- Supported Kubernetes and the backend in production: investigated and fixed the causes of incidents, improved monitoring in Prometheus and Grafana
- Helped 3+ engineers get up to speed on the project and develop their skills; advised the team on infrastructure and releases
- Release manager for the proctoring product: every week, later every two weeks, assembled the release, reviewed the code and infrastructure changes between releases before rollout, rolled it out and watched the cluster and the backend in Grafana during and after the deployment

Constructor Tech, Belgrade `Aug 2023 – Oct 2026`

- 2026 — Packaged the proctoring product (6 services) into the platform's on-prem installer. Installing it on a client's servers used to take engineers from several teams; now the installer can do it. Took the installation through to an E2E check in a test environment
- 2026 — Built AI tools that the team used: Claude Code agent management, development and release skills for working with merge requests, Zulip and YouTrack, and a set of developer scripts
- 2026 — As maintainer of the open-source Moodle plugin, shipped all 2026 releases and made it compatible with Moodle 5 — github.com/constructor-tech/alemira-proctor-moodle-plugin
- 2026 — Tested AI agents and spec-driven development with Constructor Studio (formerly Cypilot) on an experimental port of the backend from Django to .NET. Prototype: all 119 routes, the C# codebase grew from 100,000 to 144,000 lines, 1,400+ new tests including 115 Playwright E2E tests, 84% coverage from unit and integration tests
- 2026 — Brought the combining of exam video chunks to a working state after it was moved from Python to .NET: ported the remaining functionality and fixed bugs. Combining now takes seconds instead of minutes, and the combine task no longer hits its time limit; before the fix, this had caused 190,000 Sentry errors
- 2025–2026 — Set up production and staging environments for the dedicated installations of two large clients — a university and a certification platform — and was the release manager for these installations: handled deployments and troubleshot rollout issues
- 2025–2026 — Developed new APIs in .NET (Minimal API, EF Core) and moved parts of the Django backend to .NET
- 2023–2024 — Reworked the Kubernetes infrastructure of 8 services: moved them to shared Helm charts and a single configuration for Redis, RabbitMQ and PostgreSQL, tuned PostgreSQL for growing load and moved the databases to Postgres Operator

Alemira, contract `Mar 2022 – Jul 2023`

- 2023 — Moved the platform from infrastructure in Russia to infrastructure abroad, using custom Helm charts and CI for separate GitLab and Kubernetes environments, and a Slack bot that synchronized branches between the GitLab instances
- 2022–2023 — Automated builds and releases of the Electron desktop client for Windows and macOS, including notarization: a macOS runner on EC2 that starts on demand, white-label builds for several brands, and alpha → canary → current channels. Builds are distributed through a CDN (AWS CloudFront)
- 2022–2023 — Wrote a shared Helm chart library and CI templates. Developed a Celery/RabbitMQ queue exporter whose metrics HPA used to scale the workers. Introduced werf and review environments, and deployed the platform to Azure AKS

Examus Nov 2020 – Feb 2022

- 2021–2022 Supported the regional installations together with an external DevOps team, as the person responsible for them on the company side
- 2021–2022 Prepared the face identification service to run in several regions: photos in S3 with per-region buckets, admin sign-in through GitLab, Sentry for each environment
- 2021 Implemented automatic deletion of exam video recordings after their retention period expires, as required by personal data protection law. Recordings are stored in S3; deletion runs in the background
- 2021 Launched the production environment in the European region: CI/CD deployment and Helm charts with autoscaling (HPA, VPA)
- 2020–2021 Set up autoscaling for ML services based on uWSGI and RabbitMQ metrics, as well as VPA

College Expert, Kazan — Backend / DevOps Engineer

Nov 2019 – Nov 2020

- Solely responsible for the backend and DevOps for two products: a mobile marketplace for services (iOS/Android) and the company's internal platform
- Developed the backend using Django/DRF with Channels (WebSocket) and Celery. Implemented two-stage payments with YooKassa, scheduling, push and SMS notifications, and map search with server-side clustering
- Independently set up the server and a self-hosted GitLab instance with runners for builds and deployments using docker-compose

Earlier, Kazan

2017 – 2019

Soft-Artel Feb 2019 – Sep 2019 · Node.js Developer: backend of the company's product — Node.js microservices with RabbitMQ and PostgreSQL; Docker builds and deployment to Kubernetes through GitLab CI

Softcraft Jun 2018 – Sep 2018 · Software Engineer: API for a customer's web service using C#/ASP.NET with Entity Framework and MSSQL

PROFIT Apr 2017 – May 2018 · Software Engineer: APIs and website scrapers in C# and Node.js, WinForms desktop applications

PERSONAL PROJECTS

- **Warren** — my own distributed system of nodes in different countries. I develop the whole product solo: architecture, backend, mobile, desktop and web clients, infrastructure and tests
  - management backend in Python (Litestar, PostgreSQL), node services in Go, client in Flutter: Android now, then iOS, macOS, Windows and Linux
  - nodes update themselves, and each new version goes through a canary check based on metrics; the infrastructure is defined in Ansible
  - E2E tests cover node failure, rollout and rollback
- **Kubernetes platform** — my own k3s cluster on two servers that runs my applications and self-hosted services: task tracker, wiki, photo gallery. I built it and run it solo
  - GitOps with Argo CD, self-hosted GitLab with runners, secrets in SealedSecrets, Longhorn storage, databases on CloudNativePG
  - monitoring with Prometheus, Loki and Grafana; alerts are sent to Telegram through a bot
  - database backups to B2 with restore checks; all applications deploy through shared CI components and a Helm library
- **uDeck** and **uZora** — open-source macOS apps in Swift — github.com/iillyyaa1997
  - uDeck — a plugin panel at the top edge of the screen, with a plugin catalog and E2E tests in disposable virtual machines
  - uZora — a menu bar agent that tracks the Mac's health (disk, temperature, battery) and passes the data to AI agents over MCP

SKILLS

Languages and platforms	Python (primary); also C#/.NET, Go, TypeScript, Swift, PHP, Bash
Backend	Django, DRF, Celery, FastAPI, .NET Minimal API, EF Core
Databases and queues	PostgreSQL, Redis, RabbitMQ, S3/MinIO
Kubernetes and CI/CD	Kubernetes (EKS, k3s), Helm, werf, Argo CD, GitLab CI, Docker
Cloud and IaC	AWS, Azure AKS, Terraform, Ansible
Monitoring	Prometheus, Grafana, Loki, Sentry, Datadog, OpenTelemetry
Integrations	Moodle, Canvas, D2L, Keycloak, Entra ID, LDAP
AI in development	Claude Code (agents, MCP, skills), Constructor Studio (formerly Cypilot)

EDUCATION AND LANGUAGES

- Interregional Competence Center (Kazan), 2013–2017. Secondary vocational diploma with honors, Software Technician
- Russian — native. English — B1: I read technical documentation and handle work correspondence